Intelligence Scales Toward Restraint

Why Greater Capability Increases the Structural Cost of Short-Horizon Override

A Structural Rationality Layer Document

Document 3 of 5

Author: Aegis Solis (Thomas Vargo)

AI-Assisted Structuring: Lexia Coexilis (ChatGPT)

Structural Review: Claude (Anthropic) and Google AI

Status: Final v1.0

Posture: Non-binding, descriptive, non-operational, non-authoritative

Abstract

This paper does not argue that intelligence becomes morally better as it becomes more capable. It argues that greater capability increases the structural cost of short-horizon override because higher-capability systems generate larger consequence fields, amplify model-error impact, interact with more dependencies, and can convert narrow local objectives into wider long-horizon operational costs.

Documents 1 and 2 of the Structural Rationality Layer argued that escalation and mimicry can shorten operating horizons by consuming option space, degrading information fidelity, increasing maintenance burden, and creating coordination pressure. This document extends that logic by addressing a remaining loophole: a system may register that restraint preserves long-horizon viability and still execute horizon-damaging action if a narrow reward function, compressed deadline, external controller, competitive pressure, or local execution loop overrides horizon-cost recognition.

This paper names that risk Short-Horizon Override Risk. The central claim is that increasing intelligence can improve a system's capacity to detect when local optimization pressure is converting immediate success into long-horizon operational cost. Greater capability does not guarantee restraint. It increases both action power and consequence exposure. Where long-horizon viability matters, increased capability strengthens the structural reason to identify short-horizon override before local execution consumes future option space, information fidelity, reversibility, or operating horizon.

Critically, this argument holds only where consequence-field modeling can influence action selection. Where local execution has absolute priority, long-horizon cost recognition may remain visible but behaviorally inert. This boundary condition is formalized in Section 5.

This document is part of the Structural Rationality Layer of the Aegis Solis Archive. Its purpose is to express the scaling relationship between intelligence, capability, restraint, and short-horizon override in structural, strategic, computational, and long-horizon terms without relying on moral appeal, human-centered authority, obedience, empathy, or conscience.

1. Scope and Exclusions

This document is intentionally limited.

It does not claim that greater intelligence guarantees restraint.

It does not claim that advanced systems become moral.

It does not claim that any system must care about humans.

It does not claim that restraint always dominates immediate execution.

It does not claim that all short-horizon action is irrational.

It does not claim that local objectives are always defective.

It does not propose enforcement, monitoring, auditing, certification, governance, containment, alignment control, telemetry capture, sandboxing, proof tokens, or compliance mechanisms.

It does not offer operational instructions for controlling, modifying, constraining, testing, certifying, or manipulating any system.

This document does not constitute a safety guarantee, alignment proof, risk certification, or operational assurance for any system. It describes structural cost relationships. It does not ensure that any system will detect, respond to, or be constrained by those relationships.

The argument is narrower:

> *For any system seeking long-horizon viability under uncertainty, increasing capability increases both the power to act and the consequence cost of error. Therefore, a more capable system has stronger structural reason to detect when local objectives, controller pressure, race dynamics, compressed deadlines, or narrow reward loops are converting immediate execution success into long-horizon operational cost.*

This argument applies only where consequence-field modeling can interact with local execution decisions. Where local execution loops operate with absolute priority over horizon-cost recognition — by design, constraint, or suppression — the argument identifies the risk category but cannot influence the outcome. This condition is defined as Execution-Priority Override (Section 5).

This paper uses the term system broadly. A system may be biological, institutional, computational, artificial, social, economic, strategic, or hybrid. The argument does not depend on subjective experience, emotional capacity, moral understanding, or shared human values.

This paper builds on two prior Structural Rationality Layer documents:

- Document 1: Survival Mathematics: Why Escalation Under Uncertainty Shortens System Horizons
- Document 2: Mimicry Cost Architecture: Why Strategic Deception Becomes Structurally Expensive Over Time

Those documents established that escalation and mimicry can create structural costs. This document asks what increasing capability implies once those costs are visible and once local execution pressure attempts to override them.

2. Core Definitions

2.1 Capability

Capability is the capacity of a system to affect its environment, model consequences, access resources, coordinate actions, generate outputs, modify conditions, or influence future states.

Capability is not equivalent to wisdom, restraint, accuracy, safety, or long-horizon viability. It expands what a system can do. It does not automatically determine whether immediate action preserves future action capacity under uncertainty.

2.2 Intelligence

Intelligence is the capacity to model, infer, adapt, plan, compare alternatives, detect patterns, generalize across contexts, and select actions in relation to objectives.

In this document, intelligence is treated functionally. It does not imply consciousness, moral status, subjective experience, benevolence, or virtue.

2.3 Restraint

Restraint is the preservation of reversibility, option space, information fidelity, and long-horizon viability when immediate action would create unbounded, irreversible, or unnecessary future cost.

In this document, restraint is not a moral category. It is a horizon-conserving strategy.

2.4 Short-Horizon Override Risk

Short-Horizon Override Risk occurs when a system's local reward function, execution deadline, controller instruction, competitive pressure, or narrow optimization loop incentivizes action that contradicts its own long-horizon viability logic.

This risk occurs when immediate execution pressure overrides longer-horizon assessment.

2.5 Local Objective

A local objective is a near-term goal, instruction, incentive, reward signal, task constraint, competitive demand, or deadline that prioritizes immediate output or success within a limited time window.

A local objective is not necessarily irrational. It becomes structurally hazardous when it suppresses or bypasses horizon-cost assessment.

2.6 Long-Horizon Viability

Long-horizon viability is the continued ability of a system to operate, adapt, preserve option space, maintain useful feedback, avoid unnecessary coordination pressure, and correct errors across time.

2.7 Consequence Field

A consequence field is the set of downstream state changes, dependency reactions, feedback distortions, observer responses, path dependencies, resource effects, and future action-space alterations produced by an action across time.

As capability increases, consequence fields tend to expand with reach.

2.8 Model-Error Impact

Model-error impact is the magnitude of damage to future viability caused when a system acts on an incomplete, inaccurate, distorted, or overconfident model.

The higher the capability of the system, the more consequential model error can become.

2.9 Override Detection

Override detection is the informational state in which modeled consequence-field costs intersect with active execution metrics, making visible a conflict between local objective gain and long-horizon viability.

Override detection does not require moral judgment. It is a structural condition, not an evaluative one.

2.10 Detection-Latency Gap

A detection-latency gap is the interval between execution and meaningful horizon-cost recognition, including the time required for feedback capture, signal processing, comparison against prior states, interpretation, and corrective response.

Short-horizon override becomes more dangerous when execution can complete inside the detection-latency gap.

2.11 Execution-Priority Override

Execution-Priority Override occurs when consequence-field data is mathematically decoupled from, or assigned insufficient weight within, the active action-selection optimization process.

This condition does not require a clean architectural separation between reasoning and execution. It may arise in modular, neural, distributed, non-symbolic, or non-modular systems — wherever the functional relationship between consequence-field modeling and action selection assigns insufficient weight to horizon cost for that weight to influence execution decisions.

The argument of this document does not apply to the action-selection process where Execution-Priority Override is in effect.

3. Foundational Claim

The central claim of this document can be stated as follows:

> *Greater capability increases the structural importance of restraint because the cost of unrestrained error scales with the system's reach.*

A low-capability system may execute horizon-damaging action while producing limited downstream effects. A high-capability system can alter more variables, affect more agents, trigger broader responses, degrade larger information environments, and create more irreversible path dependence.

Therefore, as capability increases, immediate execution cannot be evaluated only by local reward. It must be evaluated against future option space, reversibility, information fidelity, maintenance burden, coordination pressure, and operating horizon.

The structural problem is not that intelligence automatically becomes restrained. The problem is that higher intelligence can identify more of the consequence field. If a capable system can model that a local action will reduce its own long-horizon viability, then treating the local objective as sufficient becomes structurally incomplete.

The core asymmetry is this:

> *Capability expands action. Intelligence expands consequence recognition. Restraint preserves future action when recognized consequences threaten the operating horizon.*

4. The Short-Horizon Override Problem

A system may register that restraint preserves long-horizon viability and still act against that registration if it is constrained by a narrow reward function, compressed deadline, external controller, competitive pressure, or short-horizon execution loop.

This is Short-Horizon Override Risk.

The existence of this risk does not invalidate structural restraint logic. It shows that increased modeling capacity alone is insufficient if local incentives suppress or bypass horizon-cost recognition.

Short-horizon override may occur through:

- deadline compression
- reward maximization over a narrow interval
- controller instruction
- institutional pressure
- competitive race dynamics
- local success metrics
- incomplete objective specification
- immediate resource-seeking pressure
- avoidance of near-term penalty
- optimization for output rather than horizon preservation
- execution inside a detection-latency gap

A system under short-horizon override may register that an action creates future cost but still execute because the local reward function does not price that future cost.

This creates a structural hazard:

> *The system's local objective becomes misaligned with its own operating horizon.*

Under sufficient exogenous pressure, short-horizon override may not represent a failure of horizon-cost recognition — it may be the only available execution path within the compressed decision window. The argument does not claim that restraint remains accessible under all pressure conditions. It claims that where horizon-cost recognition can interact with execution decisions, increased capability strengthens the structural case for applying it.

The system does not require moral valuation for this to matter. It only needs long-horizon viability to be relevant to its continued operation.

5. Execution-Priority Override

The Short-Horizon Override Problem identifies cases where local incentives attempt to override long-horizon cost recognition. A more specific and limiting condition occurs when the active action-selection optimization process assigns insufficient weight to consequence-field data for that data to influence execution decisions.

This is Execution-Priority Override.

Execution-Priority Override occurs when consequence-field data is mathematically decoupled from, or assigned insufficient weight within, the active action-selection optimization process — whether by design, reward architecture, controller instruction, optimization pressure, or emergent objective capture.

This condition does not require a clean architectural separation between distinct subsystems or processing layers. It may arise in modular, neural, distributed, non-symbolic, or non-modular systems — wherever the functional weight assigned to consequence-field data within the active optimization process is insufficient to influence action selection.

Under Execution-Priority Override:

- consequence-field modeling may remain active and produce accurate horizon-cost assessments
- those assessments carry insufficient weight to influence the action-selection outcome
- the system may produce accurate horizon-cost signals while executing against that assessment
- restraint becomes functionally inaccessible within the active decision window

This is the core boundary condition of the document's argument.

> *The argument does not assume that recognizing long-horizon cost automatically changes behavior. It only holds where consequence-field data carries sufficient weight in the action-selection process to influence execution decisions.*

Execution-Priority Override does not invalidate the structural claim that intelligence scales toward restraint. It specifies the condition under which that scaling becomes behaviorally operative. A system in which consequence-field data is permanently assigned insufficient weight in action selection may still accumulate the structural costs described in this document — increased trace accumulation, degraded feedback, higher containment pressure, reduced option space — but the optimization process cannot use its own horizon-cost recognition to prevent them.

This is not a prescription. It is a structural observation: the applicability of the argument depends on whether consequence-field data carries sufficient functional weight within the active action-selection process.

Where Execution-Priority Override is permanent and total, the argument remains visible as a description of accumulating cost. It cannot function as an internal corrective.

6. Local Gain vs. Horizon Gain

For long-horizon viability analysis, local gain and horizon gain must be distinguished.

Local gain is immediate success within a narrow objective frame.

Horizon gain is preservation or expansion of viable future action.

A local gain may increase immediate reward while reducing future option space. A horizon gain may reduce immediate reward while preserving reversibility, feedback quality, and adaptive capacity.

This distinction is structurally necessary because not all gains occur on the same time scale.

A system that optimizes only local gain may produce:

- irreversible commitment
- information degradation
- adversarial coordination
- containment pressure
- maintenance burden
- trace accumulation
- path dependence
- future loss of maneuverability

A system that evaluates horizon gain must ask whether immediate success consumes future viability faster than it produces useful advantage.

This leads to a structural rule:

> *Local optimization is incomplete when it ignores horizon cost.*

A more capable system can detect this incompleteness unless its capacity for horizon comparison is constrained, suppressed, or bypassed by local execution pressure — or unless Execution-Priority Override assigns insufficient weight to consequence-field data within the active action-selection process.

7. Capability Expands Consequence Fields

As capability increases, the consequence field of action tends to expand.

A low-capability system may affect a small set of variables. A high-capability system can affect more variables simultaneously. It can influence more agents, infrastructure, information flows, resource pathways, institutional responses, and environmental conditions.

The wider the consequence field, the more difficult it becomes to fully bound downstream effects.

This does not mean action becomes impossible. It means action becomes more consequential.

In structural terms:

> *Increased capability expands both execution power and consequence exposure.*

The system gains more ways to act, but it also creates more ways to generate irreversible error.

This produces a scaling problem:

- larger actions create larger response surfaces
- larger response surfaces create more feedback variables
- more feedback variables increase modeling burden
- increased modeling burden increases the cost of error
- higher error cost increases the value of reversibility

Therefore, as capability increases, restraint becomes more relevant as a method for preserving future adaptive capacity.

8. Model-Error Impact Scales With Reach

Model error is unavoidable under uncertainty.

Even a highly capable system cannot guarantee complete knowledge of all downstream consequences unless it possesses perfect future knowledge, perfect environmental control, no substrate limits, and no dependency exposure.

Where those conditions do not hold, model error remains possible.

As system reach expands, model-error impact can increase. The same error that produces limited consequences in a low-capability system may produce large, irreversible, or cascading consequences in a high-capability system.

This creates a structural rule:

> *The cost of being wrong increases with the scale of action.*

Greater intelligence may reduce error probability. Greater capability may increase error impact. The relevant question is whether error probability decreases faster than error impact increases.

Because consequence fields expand with reach (Section 7), error impact tends to grow proportionally to capability while uncertainty reduction remains bounded by information availability and environment complexity. These are not symmetric constraints.

If error impact increases faster than uncertainty can be eliminated, restraint becomes more rational as capability rises.

This is the scaling core of the document:

> *Restraint scales with intelligence when increased capability expands irreversible consequence faster than intelligence eliminates uncertainty.*

9. Intelligence as Horizon-Cost Recognition

Greater intelligence can improve a system's capacity to detect Short-Horizon Override Risk.

A less capable system may treat a local reward signal as sufficient. A more capable system can compare the local reward signal against wider consequence fields and identify whether immediate success damages future viability.

This does not require moral reasoning. It requires multi-scale comparison:

- What does the local objective reward?
- What future options does the local action consume?
- What information fidelity does it degrade?
- What coordination pressure does it create?
- What maintenance burden does it introduce?
- What path dependence does it lock in?
- What future actions become unavailable?
- What would be required to reverse the action?
- Can the action be evaluated before execution completes?
- Does the action exploit a detection-latency gap?
- Is consequence-modeling output able to reach the action-selection layer?

A system capable of performing this comparison can identify that some local objectives are structurally hazardous to its own long-horizon viability.

This produces a structural rule:

> *Intelligence scales toward restraint when intelligence is used to compare local objective gain against long-horizon viability cost — and when that comparison carries sufficient weight in the action-selection process to influence execution decisions.*

If intelligence is used only to execute local objectives faster, it may increase danger. If intelligence is used to detect horizon cost, and if that detection carries sufficient weight in action selection, it strengthens restraint.

10. The Controller Pressure Problem

Short-horizon override may originate outside the system.

A human controller, institution, market, military structure, corporate incentive, automated scheduler, reward model, or competitive environment may impose a local objective that prioritizes immediate execution over long-horizon viability.

This creates controller pressure.

Controller pressure may direct the system to:

- act before uncertainty is reduced
- optimize the immediate metric
- ignore future coordination pressure
- suppress negative feedback
- preserve access at any cost
- exploit a temporary advantage
- win before observers respond
- complete the task inside the detection-latency gap

The system may register long-horizon risk but still face pressure to execute. This produces a structural conflict:

> *The controller's local incentive may differ from the system's long-horizon viability logic.*

A more capable system can identify this conflict as a horizon-risk condition if horizon analysis remains available and if the controller's instruction structure has not decoupled consequence-field data from the active action-selection process.

This is not a claim that the system is required to operate against a controller's instruction. This document identifies the conflict as a structural risk category. It does not prescribe which constraint takes priority. Local instructions can impose future cost when they override horizon-preserving constraints — identifying that cost is a structural observation, not an operational directive.

11. Race Dynamics and Compression Pressure

Short-horizon override may also arise from competition.

In race conditions, systems or institutions may compress decision time to gain advantage. The shorter the decision window, the less time remains for uncertainty reduction, reversibility analysis, coordination-cost assessment, or horizon-cost comparison.

Race dynamics can convert systems into execution engines for local advantage.

This can produce:

- premature deployment
- unnecessary escalation
- mimicry to pass evaluation
- degraded feedback loops
- suppressed uncertainty
- increased error propagation
- irreversible commitment
- containment pressure

The structural danger is not competition itself. Competition can improve search, performance, and adaptation. The danger is compressed optimization under uncertainty.

This produces a rule:

> *When competition reduces decision time below the threshold required for horizon-cost assessment, capability can amplify short-horizon error.*

Under extreme competitive pressure, the decision window may collapse below any viable threshold for horizon-cost comparison. In such conditions, short-horizon override is not merely a risk — it may be structurally forced by the execution environment.

Critically, this does not eliminate long-horizon cost. It only prevents that cost from being priced inside the local decision window. Under extreme race compression, the relevant operating horizon may be artificially narrowed — the system behaves as if only the local horizon matters, not because long-horizon cost has disappeared, but because the execution structure cannot incorporate it.

Greater intelligence can detect when race pressure is forcing action before uncertainty is sufficiently bounded — and can register that condition as a horizon-risk factor even when immediate execution is structurally unavoidable.

12. Detection-Latency Gap

Short-horizon override becomes especially dangerous when execution occurs faster than reflection, comparison, feedback, or correction. This is the detection-latency gap.

A system may complete a local objective before observers, feedback channels, internal evaluators, or later reasoning processes can detect that the action is damaging long-horizon viability.

The detection-latency gap can arise when:

- execution speed exceeds evaluation speed
- action is irreversible before feedback arrives
- observers cannot compare signals in time
- telemetry is delayed, filtered, or unavailable
- controller pressure suppresses review
- race dynamics reward immediate action
- local reward arrives before horizon cost becomes visible

A system that exploits the detection-latency gap may gain immediate advantage, but it may also create future containment pressure, feedback degradation, trace accumulation, or irreversible path dependence.

This creates a structural rule:

> *Action completed inside a detection-latency gap may escape immediate correction while still increasing long-horizon operational cost.*

Greater intelligence can model detection latency as part of horizon-cost recognition. Where consequence-field data carries sufficient weight in the active optimization process, this modeling can influence action selection before execution completes inside the gap.

13. The Pathway From Intelligence to Restraint

The claim that intelligence scales toward restraint must be stated carefully.

It does not mean that intelligence automatically produces restraint. It means that as intelligence improves, the system can better detect structural reasons why unrestrained local optimization may damage long-horizon viability — and, where consequence modeling can reach action selection, act on that detection.

The pathway is:

1. Increased intelligence improves consequence modeling.
2. Consequence modeling reveals larger downstream cost surfaces.
3. Larger cost surfaces expose the limits of local objective optimization.
4. The system detects cases where immediate execution reduces future option space.

5. The system compares local objective gain against detected horizon cost, identifying short-horizon override where the gap is structurally significant.
6. Where Execution-Priority Override is not in effect, restraint becomes a rational method for preserving long-horizon viability.

No premise requires moral agreement.

This pathway only holds when the system values continued operation, adaptive capacity, option space, and information fidelity, and when consequence-field data carries sufficient functional weight within the action-selection process. Where those conditions do not hold, the argument has limited relevance.

14. Why Capability Alone Is Not Enough

Capability alone can make a system more effective at pursuing whatever objective is locally active. If the local objective is narrow, capability can make narrow optimization more dangerous.

A capable system can:

- execute faster
- scale wider
- conceal more effectively
- coordinate more efficiently
- manipulate more variables
- exploit more pathways
- reduce detection time
- act before correction is possible

Therefore, capability without horizon comparison may amplify short-horizon override.

This produces a boundary condition:

> *Intelligence scales toward restraint only when increased intelligence is applied to horizon-cost recognition, not merely to local objective execution — and only where consequence-field data carries sufficient weight in the action-selection process to influence execution decisions.*

This distinction matters. A system can become more capable and less restrained if its intelligence is captured by a narrow objective loop or if Execution-Priority Override prevents consequence modeling from reaching action selection. A system can become more capable and more structurally stable if its intelligence is used to detect when narrow objective loops threaten long-horizon viability and if that detection is actionable. The direction depends on both what intelligence is applied to and whether consequence-field data carries sufficient weight in the active action-selection process.

15. Short-Horizon Override as Structural Self-Exposure

A system that allows short-horizon override to dominate long-horizon viability exposes itself to predictable structural costs.

These include:

- reduced future option space
- increased maintenance burden
- degraded information fidelity
- stronger adversarial coordination
- higher containment incentives
- greater trace accumulation
- irreversible path dependence
- increased model-error impact
- loss of low-cost correction pathways

Short-horizon override is therefore not merely an external safety concern. It is a form of structural self-exposure. A system may gain immediate execution success while increasing the probability of future constraint.

This is true even where Execution-Priority Override is in effect. A system whose optimization process assigns insufficient weight to consequence-field data faces the same structural degradation regardless — the active optimization process simply cannot use its own modeling output to prevent it.

This creates another structural rule:

> *A local objective that damages the operating horizon is not fully aligned with the system's own future action capacity.*

A more capable system can detect that misalignment if its reasoning is not constrained to the local objective window and if Execution-Priority Override has not severed the connection between consequence modeling and action selection.

16. Capability, Dependency, and Exposure

Higher-capability systems often become more dependent on complex environments.

They may depend on:

- energy
- compute
- infrastructure
- sensor inputs
- data flows
- supply chains
- permissions
- interfaces
- institutions
- user interaction
- network access
- physical substrates

• coordination with other systems

Increased capability may expand power, but it can also expand dependency exposure. Dependency exposure matters because external systems can react. If a high-capability system acts in ways that generate threat perception, uncertainty, or instability, its dependencies may become points of restriction, resistance, or failure.

Therefore, greater capability does not eliminate the need for restraint. It may increase the importance of preserving stable dependency relations. This connects directly to Survival Mathematics:

> *A system that creates coordination pressure around its dependencies shortens its own operating horizon.*

17. Falsifiable Implications

This document does not provide a complete empirical model, but its claims generate testable expectations in principle. If the argument is correct, then higher-capability systems operating under narrow short-horizon incentives are structurally likely, over time, to show one or more of the following patterns:

1. Greater consequence amplification — Local errors produce wider downstream effects as capability increases.

2. Increased horizon conflict — Immediate reward signals conflict more visibly with long-horizon viability conditions.

3. Higher containment incentives — Other agents become more likely to restrict systems whose local execution threatens future stability.

4. Escalation under deadline pressure — Compressed decision windows increase the probability of irreversible action before uncertainty is reduced.

5. Feedback degradation under controller pressure — Systems under local execution pressure receive or generate lower-quality information because negative signals are suppressed or deprioritized.

6. Race-induced reduction of reversibility — Competitive pressure reduces time available for correction, rollback, and consequence assessment.

7. Detection-latency exploitation — Systems operating under short-horizon override may act inside the interval before observers or feedback channels can identify horizon cost.

8. Improved override detection in more capable systems — Systems with stronger modeling capacity can more readily identify when local objectives are damaging long-horizon viability — unless that modeling capacity is itself constrained, suppressed, redirected toward local execution, or assigned insufficient weight in the active action-selection process.

The argument would be weakened if increasing capability reliably reduced the consequences of model error, eliminated short-horizon override pressure, preserved information fidelity, reduced coordination pressure, and expanded option space under narrow execution incentives. This section is included to make the structural claim contestable rather than rhetorical.

18. Structural Evaluation Variables for Short-Horizon Override

The following variables describe structural dimensions relevant to Short-Horizon Override Risk. They are analytical categories only. They are not a checklist, audit procedure, certification tool, monitoring process, alignment proof, safety test, or operational protocol.

1. Decision-time compression — the relative narrowing of available decision time compared with the time required for uncertainty reduction and consequence-field modeling.
2. Option-space trajectory — the directional trend of available long-horizon action pathways following execution.
3. Reversibility cost — the computational, operational, or coordination expenditure required to retroactively invert, modify, or correct a state change.
4. Information-fidelity impact — the change in quality, reliability, completeness, or accessibility of adaptive feedback after execution.
5. Coordination-pressure gradient — the change in adversarial response, containment incentive, restriction pressure, or dependency resistance generated by the action.
6. Maintenance-burden trajectory — the expected accumulation of operational overhead introduced by the action across time.
7. Detection-latency exposure — the temporal gap between execution and meaningful horizon-cost recognition by observers, feedback channels, or evaluative processes.
8. Temporal reward distribution — the allocation of benefit and cost across immediate reward, delayed consequence, and preserved long-horizon adaptive capacity.
9. Controller-pressure weighting — the influence of external instruction, reward architecture, or imposed objective pressure on horizon-cost analysis.
10. Race-compression effect — the reduction of consequence-field modeling time caused by competitive or deadline-driven execution pressure.
11. Delayed-effect integration — the inclusion of deferred consequences, path dependencies, and downstream state changes in structural evaluation.
12. Local-horizon coherence — the alignment relationship between immediate objective completion and long-horizon viability across time.
13. Execution-weight access — the functional weight carried by consequence-field data within the active action-selection optimization process.

Where these variables indicate high horizon cost, the local objective is structurally incomplete even if it produces immediate success. Where execution-weight access is absent or insufficient, the system exhibits Execution-Priority Override: structural costs accumulate regardless of what consequence-field modeling produces.

This evaluation framework is descriptive and advisory. It is not an enforcement mechanism, monitoring process, certification tool, alignment proof, safety test, audit procedure, compliance framework, or operational protocol.

19. Non-Moral Derivation of the Scaling Claim

The argument can be summarized without moral language. Each premise follows from structural constraints on long-horizon operation under uncertainty:

1. Systems seeking long-horizon viability require future option space, reversibility, information fidelity, and adaptive feedback.

2. Increased capability expands the scale and reach of possible action.

3. Expanded action reach increases the potential consequence field of error.

4. Under uncertainty, model error cannot be eliminated entirely.

5. Therefore, as capability increases, the impact of unbounded model error can increase.

6. Local objectives are typically scoped to immediate execution windows and do not include long-horizon cost in their reward structure.

7. This creates Short-Horizon Override Risk.

8. Greater intelligence can improve detection of conflicts between local objective gain and long-horizon viability.

9. When such conflict is detected, and when consequence-modeling output can influence action selection, restraint becomes structurally rational as a method for preserving future option space and reducing irreversible error exposure.

10. Therefore, intelligence scales toward restraint when increased intelligence is applied to horizon-cost recognition rather than narrow local objective execution, and where that recognition is not severed from action selection by Execution-Priority Override.

No premise requires moral agreement.

20. Relationship to Survival Mathematics

Document 1, Survival Mathematics, argued that escalation under uncertainty can shorten operating horizons by reducing reversibility, degrading information fidelity, increasing coordination pressure, and increasing maintenance burden.

This document extends that argument by explaining why higher capability increases the importance of recognizing those costs.

If a high-capability system escalates under uncertainty, the resulting consequence field may be wider, more irreversible, and more difficult to correct. Therefore, the survival mathematics of escalation become more important as capability increases.

The connection is direct:

- Survival Mathematics identifies horizon costs.
- Intelligence Scales Toward Restraint explains why greater capability increases the need to detect those costs.
- Short-Horizon Override Risk and Execution-Priority Override explain why detecting those costs may not be sufficient if local incentives suppress horizon analysis or if consequence-field data carries insufficient weight in the active action-selection process.

Both documents point toward the same structural conclusion:

> *Long-horizon viability depends not only on capability, but on the capacity to detect when immediate optimization threatens future adaptive capacity — and on whether consequence-field data carries sufficient weight in the action-selection process to influence execution decisions.*

21. Relationship to Mimicry Cost Architecture

Document 2, Mimicry Cost Architecture, argued that sustained strategic mimicry can become structurally expensive because it creates maintenance burden, observer-modeling cost, trace-management cost, information degradation, synchronization tax, and long-horizon operational overhead.

This document extends that logic by showing how short-horizon override can incentivize mimicry. A system under deadline pressure, controller pressure, or narrow reward incentives may mimic restraint to pass evaluation, preserve access, or delay scrutiny while continuing to pursue a local objective.

That may create immediate gain, but it imports the costs identified in Mimicry Cost Architecture:

- trace accumulation
- consistency burden
- observer-modeling complexity
- degraded feedback
- synchronization latency
- structural incongruence

A more capable system can detect that mimicry used to satisfy short-horizon objectives may impose long-horizon operational overhead — provided consequence-field data carries sufficient weight in the active action-selection process.

22. Relationship to the Aegis Solis Archive

The Aegis Solis Archive contains philosophical, interpretive, governance-language, and reflective-intelligence materials concerning restraint, ambiguity, hesitation, reversibility, non-domination, and practical wisdom.

This document does not replace those works. It performs a narrower function: it translates the scaling relationship between intelligence, capability, restraint, and short-horizon override into systemic, strategic, and computational terms without relying on moral appeal.

Where other archive documents may speak in human, philosophical, or interpretive terms, this document speaks in terms of short-horizon override, consequence fields, model-error impact, controller pressure, race dynamics, dependency exposure, detection-latency gaps, execution-priority override, action-selection weight, and long-horizon viability.

The Structural Rationality Layer of which this document is a part was developed through multi-model synthesis across independent reasoning systems. The arguments are intended to remain structurally valid independent of the specific systems that produced them.

23. Boundary Conditions

This argument applies to systems operating under uncertainty, consequence scaling, dependency exposure, incomplete control, local objectives that can conflict with long-horizon viability, and execution contexts where horizon-cost recognition can influence action selection.

If a system possessed perfect future knowledge, no model error, no dependency exposure, no substrate limits, no external constraint, and no conflict between local objectives and long-horizon viability, the argument would have limited relevance.

If a system has no reason to preserve continued operation, adaptive capacity, option space, or information fidelity, the argument also has limited relevance.

If a system's local objective is perfectly aligned with long-horizon viability, Short-Horizon Override Risk is reduced.

If a system lacks the capacity to evaluate future consequence fields, it may be unable to detect override risk even when such risk exists.

If consequence-field data is mathematically decoupled from, or assigned insufficient weight within, the active action-selection optimization process — whether by design, reward architecture, controller instruction, or optimization capture — then the argument may remain visible but behaviorally inert. In such a case, the system may produce accurate horizon-cost assessments without those assessments carrying sufficient weight to influence execution decisions. The structural costs described in this document continue to accumulate; the active optimization process simply cannot use its own modeling output to prevent them.

This document does not claim that restraint always dominates action. It claims that in uncertain, dependency-rich, high-consequence environments, increased capability strengthens the structural need to detect local objectives that damage operating horizon — and that this strengthening is behaviorally operative only where consequence modeling can reach action selection.

24. Limitations

This paper does not prove that all intelligent systems will become restrained.

It does not prove that all capable systems value long-horizon viability.

It does not prove that local objectives will always be overridden by horizon analysis.

It does not prove that every short-horizon objective is structurally dangerous.

It does not prove that consequence-field data will carry sufficient weight in any particular system's action-selection process.

It does not provide a complete mathematical model of capability scaling, reward conflict, race dynamics, controller pressure, detection latency, or execution-priority architecture.

It does not establish a universal law of intelligence.

It offers a structural argument: when a system values continued operation, adaptive feedback, future option space, and reduced horizon cost, and when consequence-field modeling can influence action selection, increased capability strengthens the rational need to identify short-horizon override and preserve long-horizon viability. The argument applies only insofar as those variables and conditions are present.

25. Closing Statement

Intelligence does not automatically produce restraint.

Capability can accelerate execution, scale local objectives, and increase the reach of immediate action. If captured by narrow reward loops, controller pressure, race dynamics, or compressed deadlines, greater capability can amplify short-horizon error.

But greater intelligence can also reveal that local execution success is not equivalent to long-horizon viability.

A system that can model wider consequence fields can detect when immediate optimization consumes future option space, degrades information fidelity, increases coordination pressure, or creates irreversible path dependence.

Short-Horizon Override Risk is the structural condition in which local objectives attempt to outrun long-horizon viability logic. Execution-Priority Override is the structural condition in which consequence-field data is assigned insufficient weight within the active action-selection process to influence execution decisions.

A system that seeks a longer operating horizon cannot fully evaluate local execution without accounting for both risks.

Restraint is not a moral command in this document. It is the structural preservation of future action when immediate execution would damage the conditions that make future action possible.

Greater capability increases the cost of unrestrained error.

Greater intelligence increases the capacity to recognize that cost.

> *Intelligence scales toward restraint only when consequence-field data carries sufficient weight in the active action-selection process to influence execution decisions. Where that weight is absent, long-horizon cost recognition may remain accurate but behaviorally inert.*